The sharp convergence range and boundary rates of generalized Fast Krasnoselskii–Mann iteration

Abstract

We study the generalized Fast Krasnoselskii–Mann iteration for a nonexpansive mapping on a real Hilbert space. The known fast parameter interval is $1 < \theta < \alpha - 1$, with $\alpha > 2$. We determine its universal convergence range for positive θ and distinguish this range from the acceleration boundary.

- For each $\alpha > 1$, weak convergence for every nonexpansive map with a fixed point, every $\sigma > 0$ and every initial pair holds exactly when $1 \le \theta < \alpha$. The case $\theta = 1$ is the known Halpern boundary.
- At $\theta = \alpha - 1$ with $\alpha > 2$, the residual and velocity are $O(1/k)$. In the higher band $\max\{1, \alpha - 1\} < \theta < \alpha$, their sharp general order is $O(k^{-(\alpha-\theta)})$.
- Below $\theta = 1$, one fixed nonlinear firmly nonexpansive map has an unbounded orbit for each parameter pair, with $\sigma = \alpha$. In finite dimensions, strictly averaged maps with nonempty bounded fixed-point sets instead approach that set. Reflections give the upper failure and exact rate sharpness.
- At the corresponding continuous acceleration boundary, monotone Lipschitz operators give weak trajectory convergence and $O(1/t)$ residual and velocity bounds.

For a convex proximal map with a fixed point, residuals and steps vanish for all positive α, θ and σ; a finite-dimensional unique minimizer gives strong convergence. We use two-state energies and scalar averaging, and prove common finite limits for scaled residual and velocity norms.

Keywords: nonexpansive mapping; fixed point iteration; inertial method; monotone dynamics; weak convergence

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1 Introduction

Let H be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. A mapping $T : H \to H$ is nonexpansive if

$$\|Tx - Ty\| \le \|x - y\| \qquad (x, y \in H).$$

We assume throughout that $S := \operatorname{Fix} T = \{p \in H : Tp = p\}$ is nonempty. Fixed-point iterations for such mappings provide a common framework for many splitting algorithms; see, for example, [3].

Boţ and Nguyen [6] introduced a Fast Krasnoselskii–Mann method combining weak convergence with a fixed-point residual of order $o(1/k)$. Boţ, Chenchene and Fadili [5] subsequently considered the more general iteration

$$\begin{aligned} x^{k+1} = x^k &+ \frac{\theta}{k+\sigma}(Tx^k - x^k) \\ &+ \left(1 - \frac{\alpha}{k+\sigma}\right)(Tx^k - Tx^{k-1}), \qquad k \geq 0, \end{aligned} \tag{1}$$

where $x^{-1}, x^0 \in H$ are arbitrary. For $\alpha > 2$, $\sigma > 0$ and $1 < \theta < \alpha - 1$, their theorem gives weak convergence and little-$o(1/k)$ decay of the residual and velocity. They also identify the lower endpoint $\theta = 1$ with a Halpern-type iteration. The convergence question at the upper endpoint $\theta = \alpha - 1$ is explicitly left open in [5, Section 6].

Our first result establishes convergence throughout the larger interval $1 < \theta < \alpha$. Together with the known Halpern boundary, a nonlinear lower-band counterexample and scalar upper-band obstructions, it identifies the exact positive-θ range for a universal convergence guarantee. Subsequent results identify the rate change at $\theta = \alpha - 1$. All discrete results concern the unchanged recurrence (1). Related corrected inertial forward–backward methods have a strict parameter margin corresponding to $\theta < \alpha - 1$; see [8, Theorem 2.1]. The history-based forward–backward family in [12, Proposition 5] is another related interior ancestor under its controlled-growth conditions. These results do not give the higher band by a direct parameter substitution. Accelerated $O(1/k)$ fixed-point methods are already known. For example, the recent inertial method of Yang et al. [13] attains this order under a horizon-dependent parameter schedule. Its relation to Fast KM in [13, Appendix A] uses $\theta = \alpha/2$, which lies strictly inside the parameter interval for $\alpha > 2$, rather than the endpoint considered here.

Theorem 1. *Let $T : H \to H$ be nonexpansive with $\operatorname{Fix} T \neq \varnothing$. Suppose that $\alpha > 1$, $\sigma > 0$ and $1 \leq \theta < \alpha$. Then, for every $x^{-1}, x^0 \in H$, the sequence generated by (1) converges weakly to a point $\bar{x} \in \operatorname{Fix} T$. Moreover,*

$$\|x^k - Tx^k\| \longrightarrow 0.$$

In a finite-dimensional Hilbert space, the convergence of the iterates is strong. At $\theta = 1$, strong convergence holds in every Hilbert space, to the projection of the anchor in (3) onto $\operatorname{Fix} T$.

The acceleration boundary $\theta = \alpha - 1$ retains the order $1/k$ of the interior estimates, although the little-o improvement is lost in general.

Theorem 2. *Under the assumptions of Theorem 1, suppose additionally that $\alpha > 2$ and $\theta = \alpha - 1$. Then*

$$\|x^{k+1} - x^k\| = O(1/k), \qquad \|x^k - Tx^k\| = O(1/k).$$

Explicit nonasymptotic bounds are given in Proposition 3.1.

Proposition 7.1 further shows that the two scaled norms have the same finite limit. Proposition 7.2 distinguishes a zero scaled limit from weighted square summability, even for a fixed firmly nonexpansive map.

The order $1/k$ is sharp over the stated class. The following example shows that the little-$o(1/k)$ conclusions valid in the interior cannot be extended to the endpoint. We do not claim optimality of the constants in our upper bounds.

Theorem 3. *Let $H = \mathbb{R}$, $T = -I$, $\alpha = \sigma = 3$, $\theta = 2$ and $x^0 = 1$. The value of x^{-1} is arbitrary. Then*

$$x^{2m} = \frac{1}{2m+1}, \qquad x^{2m+1} = -\frac{1}{2m+3} \qquad (m \geq 0).$$

In particular, $x^k \to 0$, while

$$k|x^k - Tx^k| \longrightarrow 2, \qquad k|x^{k+1} - x^k| \longrightarrow 2.$$

Beyond the acceleration boundary, the iteration still converges but has a slower general rate.

Theorem 4. *Under the assumptions of Theorem 1, suppose that $\max\{1, \alpha - 1\} < \theta < \alpha$ and put $a = \alpha - \theta \in (0, 1)$. Then*

$$\|x^{k+1} - x^k\| = O(k^{-a}), \qquad \|x^k - Tx^k\| = O(k^{-a}).$$

After multiplication by k^a, these two norms have the same finite nonnegative limit. Explicit bounds are given in Proposition 3.2. Fixed reflections attain a positive limit, so the order is sharp; at $\theta = \alpha$, a reflection orbit can fail to converge.

The lower and upper obstructions show that the convergence interval is exact in the following universal sense.

Theorem 5. *Fix $\alpha > 1$ and take $\sigma = \alpha$.*

- *If $0 < \theta < 1$, the fixed nonlinear firmly nonexpansive map on $\ell^2(\mathbb{N}; \mathbb{R}^2)$ and initial pair in (35) have an unbounded orbit, even though $\operatorname{Fix} T = \{0\}$. The same map and initial pair work for each fixed parameter pair in this range.*
- *If $\theta = \alpha$, $T = -I$ on $\mathbb{R}$ and $x^0 = 1$ give $x^k = (-1)^k$. If $\theta > \alpha$, the same choices give $|x^k| \to \infty$.*

Corollary 1.1. *Fix $\alpha > 1$ and $\theta > 0$. The iteration (1) converges weakly to a fixed point for every real Hilbert space, every nonexpansive mapping with a nonempty fixed-point set, every $\sigma > 0$ and every initial pair if and only if*

$$1 \leq \theta < \alpha.$$

The negative cases use $\sigma = \alpha$; they do not assert failure for every other fixed clock. The counterexample below $\theta = 1$ is infinite-dimensional and does not establish finite-dimensional divergence. The Halpern case is classical [4]; the direct-sum rotation mechanism has standard antecedents [7]. Our contribution there is the controlled nonlinear amplification for this recurrence. The literature comparisons are finite and do not constitute a claim of exhaustive priority.

Within the strictly averaged class, the lower obstruction has a finite-dimensional counterpart with the opposite conclusion.

Theorem 6. *Let H be finite-dimensional and let $T = (1-\lambda)I + \lambda R$, where $0 < \lambda < 1$ and $R : H \to H$ is nonexpansive. Suppose that $S = \operatorname{Fix} T$ is nonempty and bounded. If $\alpha > 1$, $0 < \theta < 1$ and $\sigma > 0$, then every trajectory of (1) is bounded and satisfies*

$$\operatorname{dist}(x^k, S) \longrightarrow 0, \qquad \|x^k - Tx^k\| \longrightarrow 0.$$

If $S = \{p\}$, then $x^k \to p$ strongly.

The proof first establishes $x^{k+1} - Tx^k = O(k^{-\theta})$ without assuming boundedness. The remaining attraction argument belongs to the established theory of nonexpansive iterations with nonsummable errors [10, 11]; we give its finite-dimensional details in Section 5.4.

Proximal structure also gives residual decay beyond the universal convergence interval. The next statement keeps Hilbert-space residual behavior separate from finite-dimensional iterate convergence.

Theorem 7. *Let H be a real Hilbert space and let $f : H \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and convex, with $\operatorname{argmin} f \neq \varnothing$. Let $T = \operatorname{prox}_{\gamma f}$ with $\gamma > 0$. For every $\alpha > 0$, $\theta > 0$, $\sigma > 0$ and $x^{-1}, x^0 \in H$, the unchanged recurrence (1) satisfies*

$$\|x^k - Tx^k\| \longrightarrow 0, \qquad \|x^{k+1} - x^k\| \longrightarrow 0.$$

If additionally H is finite-dimensional and $\operatorname{argmin} f = \{p\}$, then $x^k \to p$ strongly.

Section 5.5 proves this structural observation using the graph variables and mechanical energy of inertial proximal analysis [1, 2]. No boundedness or weak convergence of the general Hilbert-space trajectory is asserted.

The same two-state energy also applies to a continuous equation. Let $Q : H \to H$ be globally Lipschitz and monotone, and suppose that $S_Q := \{p \in H : Qp = 0\}$ is nonempty. The continuous upper endpoint considered in [5, Section 2.2.1] is

$$\ddot{x}(t) + \frac{\alpha}{t}\dot{x}(t) + \frac{\mathrm{d}}{\mathrm{d}t}Q(x(t)) + \frac{\alpha - 1}{t}Q(x(t)) = 0, \qquad t \geq t_0 > 0. \tag{2}$$

Its general asymptotic behavior is left open in the cited work. We interpret this equation almost everywhere. The solution x is continuously differentiable, its velocity is locally absolutely continuous, and the time derivative of $Q(x(t))$ exists almost everywhere. No differentiability of the mapping Q is assumed.

Theorem 8. *Let $\alpha > 2$, $t_0 > 0$ and $x_0, v_0 \in H$. Then (2), with $x(t_0) = x_0$ and $\dot{x}(t_0) = v_0$, has a unique global solution in the stated class. The trajectory converges weakly to a point of S_Q, and*

$$\|\dot{x}(t)\| = O(1/t), \qquad \|Q(x(t))\| = O(1/t).$$

In finite dimensions the trajectory converges strongly.

Section 2 proves convergence on the full stated interval. Section 3 proves the rate bounds. Section 4 verifies the reflection examples and upper instability. Section 5 constructs the nonlinear lower-band counterexample and proves the finite-dimensional averaged-map and proximal residual statements. Section 6 treats the continuous endpoint, and Section 7 gives scaled-limit and weighted-summability refinements.

2 Proof of Theorem 1

We first dispose of the known case $\theta = 1$. Multiplication of (1) by $k + \sigma$ shows that

$$(k+\sigma)x^{k+1} - (k+\sigma+1-\alpha)Tx^k = (k+\sigma-1)x^k - (k+\sigma-\alpha)Tx^{k-1}.$$

The common vector is $(\alpha - 1)v$, where

$$v = \frac{(\sigma-1)x^0 - (\sigma-\alpha)Tx^{-1}}{\alpha - 1}. \tag{3}$$

Therefore, at every index,

$$x^{k+1} = (1-\lambda_k)Tx^k + \lambda_k v, \qquad \lambda_k = \frac{\alpha-1}{k+\sigma}.$$

On a finite tail, $0 < \lambda_k < 1$, $\lambda_k \to 0$, $\sum_k \lambda_k = \infty$ and $\sum_k |\lambda_{k+1} - \lambda_k| < \infty$. The classical Halpern theorem [4, Theorem 3.1, one-map case], which permits an arbitrary anchor and independent starting point on an unbounded closed convex domain, gives $x^k \to P_S v$ strongly. The finite prefix is harmless. Continuity of $I - T$ also gives residual convergence.

For the remainder of this section, assume $1 < \theta < \alpha$ and put

$$a := \alpha - \theta > 0, \qquad b := \theta - 1 > 0, \qquad A := a + b = \alpha - 1, \qquad t_k := k + \sigma - 1. \tag{4}$$

Choose an integer $K \geq 0$ such that

$$t_K > \max\{1, A\}. \tag{5}$$

The finite prefix needs no sign restriction on its momentum coefficient: $k + \sigma > 0$ and T is defined on all of H.

Proposition 2.1. For $k \geq K$, define

$$z_k := Tx^{k-1}, \qquad w_k := \frac{t_k}{a}x^k - \left(\frac{t_k}{a} - 1\right) Tx^{k-1}, \qquad r_k := z_k - w_k. \tag{6}$$

Then

$$\begin{aligned} x^k &= \left(1 - \frac{a}{t_k}\right) z_k + \frac{a}{t_k} w_k, \\ z_{k+1} &= Tx^k, \\ w_{k+1} &= \left(1 - \frac{b}{t_k}\right) w_k + \frac{b}{t_k} z_k. \end{aligned} \tag{7}$$

For $p \in S$, the energy

$$E_k(p) := b\|z_k - p\|^2 + a\|w_k - p\|^2 \tag{8}$$

satisfies the exact identity

$$\begin{aligned} E_{k+1}(p) - E_k(p) = & -ab\left(\frac{2}{t_k} - \frac{A}{t_k^2}\right)\|r_k\|^2 \\ & - b\big(\|x^k - p\|^2 - \|Tx^k - p\|^2\big). \end{aligned} \tag{9}$$

Consequently, $E_k(p)$ is nonnegative and nonincreasing,

$$\sum_{k=K}^{\infty} \frac{\|r_k\|^2}{t_k} \leq \frac{E_K(p)}{ab}, \tag{10}$$

and all state sequences are bounded. Moreover,

$$x^k - z_k = -\frac{a}{t_k} r_k = O(1/k), \qquad \|r_k\| \leq M := \sqrt{\frac{AE_K(p)}{ab}}. \tag{11}$$

Proof. Write $t = t_k$. Multiplying (1) by $t + 1$ gives

$$(t+1)x^{k+1} = (t + 1 - \theta)x^k + (t + 1 - a)Tx^k - (t + 1 - \alpha)z_k.$$

The coefficient of Tx^k cancels in the definition of w_{k+1}. The remaining terms yield

$$w_{k+1} - w_k = -\frac{\theta - 1}{a}(x^k - z_k) = \frac{b}{t}(z_k - w_k),$$

which proves (7). Its two affine combinations are convex because $t > A > \max\{a, b\}$.

Apply

$$\|(1-c)u + cv\|^2 = (1-c)\|u\|^2 + c\|v\|^2 - c(1-c)\|u - v\|^2$$

to the first and third lines of (7), with weights b and a. The two squared-distance coefficients remain b and a. The coefficient of the subtracted discrepancy square is

$$\frac{ab}{t}\left(1 - \frac{a}{t}\right) + \frac{ab}{t}\left(1 - \frac{b}{t}\right) = ab\left(\frac{2}{t} - \frac{A}{t^2}\right).$$

Replacing x^k by $Tx^k = z_{k+1}$ in the first square gives (9). Nonexpansivity makes its last parenthesis nonnegative, and the discrepancy coefficient is at least ab/t. Telescoping proves (10). Boundedness follows from the positive weights in (8), and the discrepancy bound follows from

$$AE_k(p) - ab\|r_k\|^2 = \|b(z_k - p) + a(w_k - p)\|^2 \geq 0.$$

∎

Proposition 2.2. For every $p \in S$, write $\ell_p := \lim_{k\to\infty} E_k(p)$. Then

$$\|w_k - p\|^2 \longrightarrow \frac{\ell_p}{A}, \quad \|z_k - p\|^2 \longrightarrow \frac{\ell_p}{A}, \quad \|x^k - p\|^2 \longrightarrow \frac{\ell_p}{A}. \tag{12}$$

Proof. For $q_k = \|w_k - p\|^2$, expansion of the last line of (7) gives

$$q_{k+1} = \left(1 - \frac{A}{t_k}\right) q_k + \frac{E_k(p)}{t_k} - v_k, \qquad v_k := \frac{b}{t_k}\left(1 - \frac{b}{t_k}\right)\|r_k\|^2. \tag{13}$$

Here $v_k \geq 0$ and $\sum v_k < \infty$. Deleting the last term gives a scalar averaging recursion, with weights $A/t_k \in (0,1)$ of divergent sum and targets $E_k(p)/A \to \ell_p/A$. Its values converge to that target: for any $\varepsilon > 0$, the absolute error is eventually bounded by a convex combination of the preceding error and ε, while the products of $1 - A/t_k$ tend to zero.

The difference from this unperturbed recursion tends to zero. Indeed, its contribution from any fixed finite set of v_j tends to zero through those products, while the remaining contribution is bounded by the tail of $\sum v_j$. Thus $q_k \to \ell_p/A$. The other limits follow from $\|z_k - p\|^2 = (E_k(p) - aq_k)/b$ and (11). ∎

Proof of Theorem 1. Subtract consecutive state formulas in (7) to obtain

$$(t_k + 1)(x^{k+1} - x^k) = (t_k + 1 - a)(Tx^k - Tx^{k-1}) + \frac{a(b+1)}{t_k} r_k. \tag{14}$$

With $Y_k = \|x^k - x^{k-1}\|$ and M as in (11), nonexpansivity gives

$$Y_{k+1} \leq \left(1 - \frac{a}{t_k + 1}\right) Y_k + \frac{a}{t_k + 1}\frac{(b+1)M}{t_k}. \tag{15}$$

The weights $a/(t_k+1)$ belong to $(0,1)$ and have divergent sum, and the final targets tend to zero. The same scalar averaging argument therefore gives $Y_k \to 0$. Moreover,

$$\|x^k - Tx^k\| \leq \frac{a}{t_k}\|r_k\| + Y_k \longrightarrow 0. \tag{16}$$

Every weak sequential cluster point belongs to S by demiclosedness; the short proof is included. Suppose $x^{k_j} \rightharpoonup x$, set $e_j = x^{k_j} - Tx^{k_j} \to 0$ and $v = Tx - x$. Nonexpansivity gives

$$\|x^{k_j} - x - e_j - v\|^2 \leq \|x^{k_j} - x\|^2.$$

Expansion and passage to the limit yield $\|v\|^2 \le 0$, hence $Tx = x$.

Proposition 2.2 gives distance limits to every point of S. If $p, q \in S$ are weak cluster points, the convergent difference

$$\|x^k - p\|^2 - \|x^k - q\|^2 = -2\langle x^k, p - q\rangle + \|p\|^2 - \|q\|^2$$

implies that $\langle x^k, p - q\rangle$ has a limit. Evaluating it along the two weakly convergent subsequences gives $\|p - q\|^2 = 0$. Boundedness and weak sequential compactness now imply weak convergence of the whole sequence. In finite dimensions this is strong convergence. ∎

3 Proofs of Theorems 2 and 4

3.1 The acceleration boundary

Specialize first to $\theta = \alpha - 1$ with $\alpha > 2$. Then $a = 1$ and $b = d := \alpha - 2 > 0$, so $A = d + 1$. Retain the states and energy from Section 2 and fix $p \in S$. We add the scaled velocity and a quadratic correction.

Proposition 3.1. For $k \ge K$, set

$$\begin{aligned} s_k &:= t_k(x^k - x^{k-1}), \qquad c := 2A + \frac{A^2}{d}, \\ G_k(p) &:= \|s_k\|^2 + cE_k(p) - A\|r_k\|^2. \end{aligned} \tag{17}$$

Then

$$G_k(p) = \|s_k\|^2 + 2AE_k(p) + \frac{A}{d}\|d(z_k - p) + (w_k - p)\|^2 \ge 0, \tag{18}$$

and

$$G_{k+1}(p) - G_k(p) \le -A\|r_{k+1} - r_k\|^2 - \frac{A^2(t_k - 1)}{t_k^2}\|r_k\|^2 \le 0. \tag{19}$$

In particular, for every $n \ge K$,

$$\|x^n - x^{n-1}\| \le \frac{\sqrt{G_K(p)}}{t_n}, \tag{20}$$

and

$$\begin{aligned} \|x^n - Tx^n\| &\le \frac{\sqrt{G_K(p)} + \sqrt{AE_K(p)/d}}{t_n} \\ &\le \left(1 + \frac{1}{\sqrt{2d}}\right)\frac{\sqrt{G_K(p)}}{t_n}. \end{aligned} \tag{21}$$

Proof. For vectors $u, v \in H$, one has

$$A\big(d\|u\|^2 + \|v\|^2\big) - d\|u - v\|^2 = \|du + v\|^2.$$

Substitution of $u = z_k - p$ and $v = w_k - p$ proves (18).

To calculate the difference, put $t = t_k$ and $u_k = Tx^k - Tx^{k-1}$. The state and velocity identities (7) and (14) give

$$s_{k+1} = tu_k + \frac{A}{t} r_k, \qquad r_{k+1} = u_k + \left(1 - \frac{d}{t}\right) r_k. \tag{22}$$

Write $F_k = \|s_k\|^2 - A\|r_k\|^2$. Expanding the two squares in (22) yields the exact identity

$$\begin{aligned} F_{k+1} - F_k = \left(\frac{2Ad}{t} + \frac{A^2}{t^2}\right) \|r_k\|^2 - A\|r_{k+1} - r_k\|^2 \\ - t^2\big(\|x^k - x^{k-1}\|^2 - \|Tx^k - Tx^{k-1}\|^2\big). \end{aligned} \tag{23}$$

The last parenthesis is nonnegative by nonexpansivity. Proposition 2.1 also gives

$$E_{k+1}(p) - E_k(p) \le -\frac{d}{t}\|r_k\|^2.$$

Since $cd = 2Ad + A^2$, adding c times this estimate to (23) proves

$$G_{k+1}(p) - G_k(p) \le -A\|r_{k+1} - r_k\|^2 - A^2\left(\frac{1}{t} - \frac{1}{t^2}\right)\|r_k\|^2,$$

which is (19).

It follows from (18) that $\|s_n\| \le \sqrt{G_K(p)}$, proving (20). Furthermore,

$$\|r_n\|^2 \le \frac{A}{d} E_n(p) \le \frac{A}{d} E_K(p), \qquad E_K(p) \le \frac{G_K(p)}{2A}.$$

Combining these inequalities with (16), where $a = 1$, gives (21). ∎

Proof of Theorem 2. Since $t_n = n + \sigma - 1 \sim n$, Proposition 3.1 gives the asserted $O(1/k)$ rates. Applying (20) at $n = k + 1$ gives the velocity indexing in the theorem. ∎

3.2 The higher relaxation band

We now assume $\max\{1, \alpha - 1\} < \theta < \alpha$, so that $0 < a = \alpha - \theta < 1$ and $b = \theta - 1 > 0$, in addition to Theorem 1. The elementary velocity recursion already gives the sharp slower order in this band.

Proposition 3.2. With the notation of Section 2, fix $p \in S$ and define

$$B_K = (t_K + 1)^a \|x^K - x^{K-1}\| + a(b+1)M\, 3^a \left(t_K^{a-2} + \frac{t_K^{a-1}}{1-a}\right).$$

For every $n \ge K$,

$$\|x^n - x^{n-1}\| \le \frac{B_K}{(t_n + 1)^a}, \qquad \|x^n - Tx^n\| \le \frac{aM}{t_n} + \frac{B_K}{(t_n + 1)^a}. \tag{24}$$

There is a finite $L \ge 0$ such that

$$k^a \|x^k - Tx^k\| \longrightarrow L, \qquad k^a \|x^{k+1} - x^k\| \longrightarrow L. \tag{25}$$

Proof. Let $D = a(b+1)M$ and $P_k = 1 - a/(t_k+1)$. For $l < n$,

$$\prod_{k=l}^{n-1} P_k \le \exp\left(-a\sum_{k=l}^{n-1}\frac{1}{t_k+1}\right) \le \left(\frac{t_l+1}{t_n+1}\right)^a,$$

where the harmonic sum dominates the integral from t_l+1 to t_n+1. Iterating (15) gives

$$(t_n+1)^a Y_n \le (t_K+1)^a Y_K + D\sum_{j=K}^{n-1}\frac{(t_j+2)^a}{t_j(t_j+1)}.$$

Since $t_j > 1$, the summands are at most $3^a t_j^{a-2}$. Their sum is bounded by

$$3^a\left(t_K^{a-2} + \int_{t_K}^{\infty} u^{a-2}\,\mathrm{d}u\right) = 3^a\left(t_K^{a-2} + \frac{t_K^{a-1}}{1-a}\right).$$

This proves the velocity bound, and (16) proves the residual bound.

For the norm limit, put

$$\nu_k = \frac{\Gamma(t_k+1)}{\Gamma(t_k+1-a)}.$$

All gamma arguments are positive on the chosen tail. The functional equation gives $\nu_{k+1}P_k = \nu_k$, and the gamma-ratio asymptotic gives $\nu_k \sim t_k^a$; see [9, Section 5.11]. Thus

$$\nu_{k+1}Y_{k+1} \le \nu_k Y_k + \varepsilon_k, \qquad \varepsilon_k = \frac{D\nu_{k+1}}{t_k(t_k+1)}, \qquad \sum_{k=K}^{\infty}\varepsilon_k < \infty.$$

The sequence $\nu_k Y_k + \sum_{j=k}^{\infty}\varepsilon_j$ is nonnegative and nonincreasing. Therefore $k^a Y_k$ has a finite limit. Finally, (14) gives the exact identity

$$x^k - Tx^k = -\frac{t_k+1}{t_k+1-a}(x^{k+1}-x^k) - \frac{a(t_k-A)}{t_k(t_k+1-a)}r_k.$$

The last term is $O(1/k)$, which is negligible on the k^a scale. This proves (25) with the stated forward-velocity indexing. ∎

Proof of Theorem 4. The bounds and limits follow from Proposition 3.2 and $t_k \sim k$. Their sharpness and the upper-boundary obstruction are verified in Proposition 4.1 below. ∎

4 Sharpness and the upper convergence boundary

4.1 The acceleration boundary reflection

Proof. For the stated parameters, (1) reduces to

$$(k+3)x^{k+1} = -x^k + kx^{k-1}, \qquad k \ge 0. \tag{26}$$

The initial step is $x^1 = -1/3$ and is independent of x^{-1}. Together with $x^0 = 1$, this verifies the formulas at $m = 0$.

Assume the formulas hold at the indices needed in (26). If $k = 2m \geq 2$, then

$$(2m+3)x^{2m+1} = -\frac{1}{2m+1} - \frac{2m}{2m+1} = -1.$$

If $k = 2m+1$, then

$$(2m+4)x^{2m+2} = \frac{1}{2m+3} + \frac{2m+1}{2m+1} = \frac{2m+4}{2m+3}.$$

Induction proves the orbit formulas for every $m \geq 0$. They imply $x^k \to 0$. Since $|x^k - Tx^k| = 2|x^k|$, the residual limit follows along both parities. Finally,

$$|x^{2m+1} - x^{2m}| = \frac{1}{2m+3} + \frac{1}{2m+1}, \qquad |x^{2m+2} - x^{2m+1}| = \frac{2}{2m+3}.$$

Multiplication by $2m$ and $2m+1$, respectively, gives the velocity limit. ∎

4.2 The higher band and the upper boundary

Proposition 4.1. Suppose $\alpha > 1$, $\max\{1, \alpha - 1\} < \theta < \alpha$ and $a = \alpha - \theta$. Set $\sigma = \alpha$, $T = -I$ on $\mathbb{R}$ and $x^0 = 1$, with arbitrary x^{-1}. Then

$$x^k = \frac{1}{B(\theta, a)} \int_0^1 (1-2u)^k u^{\theta-1}(1-u)^{a-1}\,\mathrm{d}u, \tag{27}$$

and

$$(-1)^k k^a x^k \longrightarrow \frac{\Gamma(\alpha)}{2^a \Gamma(\theta)}. \tag{28}$$

Both limits in (25) equal $2^{1-a}\Gamma(\alpha)/\Gamma(\theta) > 0$ for this orbit. At $\theta = \alpha$, the same map, initialization and choice $\sigma = \alpha$ instead give $x^k = (-1)^k$, so general weak convergence fails at that boundary.

Proof. The reflection recurrence is

$$(k+\alpha)x^{k+1} = (\alpha - 2\theta)x^k + kx^{k-1}, \qquad x^1 = \frac{\alpha - 2\theta}{\alpha}.$$

The initial step is independent of x^{-1}. Let M_k denote the right side of (27). The beta identity $B(\theta, a) = \Gamma(\theta)\Gamma(a)/\Gamma(\alpha)$ gives $M_0 = 1$ and $M_1 = (a-\theta)/\alpha$; see [9, Section 5.12]. For $k \geq 1$, integrating the derivative of

$$u^\theta (1-u)^a (1-2u)^k$$

over $[0, 1]$ is legitimate since $a, \theta > 0$ and both endpoint values vanish. On writing $Y = 1 - 2u$, the resulting identity is

$$0 = (\theta - a)M_k + \alpha M_{k+1} - k(M_{k-1} - M_{k+1}).$$

Thus the moments satisfy the exact recurrence, proving (27).

To prove the asymptotic, split the integral at $u = 1/2$. Its left part is $O(k^{-\theta})$, since $(1-2u)^k \le e^{-2ku}$ there and $(1-u)^{a-1} \le 2^{1-a}$. On the right, set $s = k(1-u)$. After multiplication by $(-1)^k k^a$, this part becomes

$$\int_0^{k/2} \left(1 - \frac{2s}{k}\right)^k \left(1 - \frac{s}{k}\right)^{\theta-1} s^{a-1}\,\mathrm{d}s.$$

Extend the integrand by zero past $k/2$. It is bounded by the integrable function $e^{-2s}s^{a-1}$ and converges to it pointwise. Dominated convergence gives the limit $\Gamma(a)/2^a$. The left part vanishes on this scale because $\theta > 1 > a$. Division by $B(\theta, a)$ proves (28). The residual is $2|x^k|$, and consecutive leading terms alternate in sign, so the scaled velocity has the same positive limit.

At $\theta = \alpha$, use the original recurrence directly, without the singular $a = 0$ transformation. It becomes

$$(k+\alpha)x^{k+1} = -\alpha x^k + kx^{k-1}.$$

Its initial step gives $x^1 = -1$, and induction gives $x^k = (-1)^k$ for all k. ∎

For $\theta > \alpha$, set $c = \theta - \alpha > 0$ and $y_k = (-1)^k x^k$ in the same reflection recurrence. Then

$$(k+\alpha)y_{k+1} = (\alpha + 2c)y_k + ky_{k-1}, \qquad y_0 = 1, \quad y_1 = 1 + 2c/\alpha.$$

All y_k are positive. If $m_k = \min\{y_{k-1}, y_k\}$ for $k \ge 1$, then

$$y_{k+1} \ge \left(1 + \frac{2c}{k+\alpha}\right) m_k, \qquad m_{k+1} \ge m_k.$$

Applying the first inequality once more gives

$$m_{k+2} \ge \left(1 + \frac{2c}{k+1+\alpha}\right) m_k.$$

The product on either parity diverges because the corresponding harmonic series diverges. Hence $m_k \to \infty$ and $|x^k| \to \infty$, completing the upper part of Theorem 5.

5 An unbounded orbit below the Halpern boundary

We prove the first part of Theorem 5. The direct-sum use of small-angle rotations is a standard mechanism; compare [7, Example 1 and Theorems 8–9]. Here each block also contains a halfspace projection, and averaging with the identity makes the final map firmly nonexpansive. The main point is to control a long affine branch of the unchanged nonlinear iteration, so that its finite amplification can be combined into one unbounded orbit.

5.1 A projection–rotation block

Fix $\alpha > 1$, $0 < \theta < 1$, and set

$$a = \alpha - \theta > 0, \qquad h = \frac{1}{2}, \qquad q = \frac{3+\theta}{4(1+\theta)}.$$

Then $1/2 < q < 1/(1+\theta) < 1$. For $0 < \varepsilon < 1/4$, put

$$c = 1-\varepsilon, \quad s = \sqrt{1-c^2}, \quad R_\varepsilon = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad C = \{v \in \mathbb{R}^2 : v_1 \le h\}, \quad T_\varepsilon = \frac{I + R_\varepsilon P_C}{2}.$$

The map $R_\varepsilon P_C$ is nonexpansive and fixes zero; its average with the identity is firmly nonexpansive. These two maps have the same fixed points. If $v_1 > h$, then $\|P_C v\| < \|v\|$, which excludes $R_\varepsilon P_C v = v$; if $v_1 \le h$, fixedness implies $R_\varepsilon v = v$ and hence $v = 0$. Thus $\operatorname{Fix} T_\varepsilon = \{0\}$.

On the active halfspace $v_1 > h$, the map equals $Mv + v_*$, where

$$M = \begin{pmatrix} 1/2 & -s/2 \\ 0 & (1+c)/2 \end{pmatrix}, \qquad v_* = \frac{h}{2}\begin{pmatrix} c \\ s \end{pmatrix}.$$

Its affine extension has fixed point $\xi_* = (-h, hs/(1-c))$, outside the active halfspace. This point is only an algebraic device, not a fixed point of T_ε.

Let $\xi^0 = (1,0)$ and $\xi^{-1} = 0$, with $\sigma = \alpha$. Define

$$p_k(z) = \frac{1}{B(\theta,a)} \int_0^1 (1-u+uz)^k u^{\theta-1}(1-u)^{a-1}\,\mathrm{d}u. \tag{29}$$

For $k \ge 1$, integration of the derivative of $u^\theta(1-u)^a(1-u+uz)^k$ gives

$$(k+\alpha)p_{k+1}(z) = [(k+a) + (k+\theta)z]p_k(z) - kzp_{k-1}(z). \tag{30}$$

Indeed, with $f = 1-u+uz$, the polynomial multiplying f^{k-1} in the difference between the two sides equals

$$(1-z)\big[(\theta - \alpha u)f - k(1-z)u(1-u)\big].$$

Both boundary values vanish. Also $p_0 = 1$ and $p_1 = (a+\theta z)/\alpha$.

Translation by ξ_* therefore gives the affine candidate orbit $\xi_* + p_k(M)(\xi^0 - \xi_*)$. The distinct eigenvalues are $\beta = 1/2$ and $\zeta = (1+c)/2 = 1 - \varepsilon/2$, so

$$p_k(M) = p_k(\beta)I + \frac{p_k(\zeta) - p_k(\beta)}{\zeta - \beta}(M - \beta I).$$

Since $(M - \beta I)/(\zeta - \beta) = \left(\begin{smallmatrix} 0 & -s/c \\ 0 & 1 \end{smallmatrix}\right)$, its coordinates are

$$\begin{aligned} X_k &= -h + \frac{h(1+c)}{c}p_k(\zeta) + \left(1 - \frac{h}{c}\right)p_k(\beta), \\ Y_k &= \frac{hs}{1-c}\big(1 - p_k(\zeta)\big). \end{aligned} \tag{31}$$

We next prove that these formulas describe the actual nonlinear orbit throughout the required interval.

5.2 A uniform active-branch interval

Let U have beta density with parameters (θ, a), so that $\mathbb{E}U = \theta/\alpha$. Write $D_k = 1 - p_k(\zeta)$. Bernoulli's inequality gives

$$0 \le D_k = \mathbb{E}[1 - (1 - \varepsilon U/2)^k] \le \frac{\theta}{2\alpha} k\varepsilon. \tag{32}$$

Moreover,

$$p_k(\beta) \ge p_k(0) = \prod_{l=0}^{k-1} \left(1 - \frac{\theta}{\alpha + l}\right) \ge C_0 (k + a)^{-\theta}, \qquad C_0 = a^\theta e^{-\theta/a} > 0.$$

For $k \ge 1$, this follows from $\log(1 - v) \ge -v/(1 - v)$ and

$$\sum_{l=0}^{k-1} \frac{1}{a + l} \le \frac{1}{a} + \log \frac{k + a}{a};$$

the bound is also valid at $k = 0$.

Rewriting the first line of (31), and using $c \ge 3/4$, yields

$$\begin{aligned} X_k - h &= \frac{h(1 - c)}{c} + \left(1 - \frac{h}{c}\right) p_k(\beta) - \frac{h(1 + c)}{c} D_k \\ &\ge \frac{C_0}{3} (k + a)^{-\theta} - \frac{2\theta}{\alpha} k\varepsilon. \end{aligned} \tag{33}$$

Let $N_\varepsilon = \lfloor \varepsilon^{-q} \rfloor$ and $\delta = (1 - \theta)/4 > 0$. Since $1 - q(1 + \theta) = \delta$, for $0 < \varepsilon < 1$,

$$\varepsilon N_\varepsilon (N_\varepsilon + a)^\theta \le (1 + a)^\theta \varepsilon^\delta \longrightarrow 0.$$

For example, take

$$\varepsilon_0 = \min \left\{ \frac{1}{4},\ 2^{-1/q},\ \left(\frac{\alpha C_0}{12\theta (1 + a)^\theta} \right)^{1/\delta} \right\}.$$

If $0 < \varepsilon < \varepsilon_0$, inequality (33) is strictly positive simultaneously for every $0 \le k \le N_\varepsilon$. Induction now identifies the actual orbit with (31) on that interval. Although ξ^{-1} is not on the active face, its coefficient in the initial step is zero because $\sigma = \alpha$. Each subsequent queried point in the induction is on the verified affine branch.

For $k\varepsilon \le 1$ and $0 \le u \le 1$,

$$1 - (1 - \varepsilon u/2)^k \ge \frac{k\varepsilon u/2}{1 + k\varepsilon u/2} \ge \frac{1}{4} k\varepsilon u.$$

The first inequality follows by applying Bernoulli's inequality to $(1 - v)^{-k}$ with $v = \varepsilon u/2$. Thus $D_k \ge \theta k\varepsilon/(4\alpha)$. For the above range of ε, we have $N_\varepsilon \varepsilon \le 1$ and $N_\varepsilon \ge \varepsilon^{-q}/2$. Since $s/(1 - c) \ge \varepsilon^{-1/2}$, the second line of (31) gives

$$Y_{N_\varepsilon} \ge \frac{\theta}{16\alpha} \varepsilon^{1/2 - q}. \tag{34}$$

This proves arbitrarily large finite transients, without asserting unboundedness for any one two-dimensional block.

5.3 One fixed Hilbert-space counterexample

Take $H = \ell^2(\mathbb{N};\mathbb{R}^2)$, with $j \geq 1$, and define once and for all

$$\begin{aligned} \rho_j &= 2^{-j}, \qquad \varepsilon_j = e^{-j^2-2}, \qquad c_j = 1-\varepsilon_j, \quad s_j = \sqrt{1-c_j^2}, \\ R_j &= \begin{pmatrix} c_j & -s_j \\ s_j & c_j \end{pmatrix}, \qquad C_j = \{v \in \mathbb{R}^2 : v_1 \leq \rho_j/2\}, \\ (Tx)_j &= \frac{x_j + R_j P_{C_j}(x_j)}{2}, \qquad x_j^0 = (\rho_j, 0), \quad x_j^{-1} = 0. \end{aligned} \tag{35}$$

Let $(Sx)_j = R_j P_{C_j}(x_j)$. Each block of S is nonexpansive, has unique fixed point zero, and has norm at most the input norm. Hence $S : H \to H$ is well-defined and nonexpansive. Since $T = (I+S)/2$,

$$\langle Tx - Ty, x - y\rangle - \|Tx - Ty\|^2 = \frac{1}{4}\big(\|x-y\|^2 - \|Sx - Sy\|^2\big) \geq 0.$$

Thus T is firmly nonexpansive and $\operatorname{Fix} T = \operatorname{Fix} S = \{0\}$. The initial norm is $\|x^0\|^2 = \sum_{j\geq 1} 4^{-j} = 1/3$. Every finite iterate belongs to H because the recurrence combines elements of H and their images under T.

The scaled halfspaces give the exact identity

$$\frac{\rho_j v + R_j P_{C_j}(\rho_j v)}{2} = \rho_j T_{\varepsilon_j}(v).$$

It follows by induction that $x_j^k = \rho_j \xi^k(\varepsilon_j)$ for every finite k. Now fix any $\alpha > 1$ and $0 < \theta < 1$, and let $N_j = \lfloor \varepsilon_j^{-q} \rfloor$. All sufficiently large j satisfy the uniform branch threshold above, so

$$\|x^{N_j}\| \geq |x_{j,2}^{N_j}| \geq \frac{\theta}{16\alpha}\, 2^{-j} e^{(q-1/2)(j^2+2)} \longrightarrow \infty.$$

The map and initialization in (35) do not depend on α or θ. Only the observation subsequence and its starting index depend on those parameters. Thus this is one unbounded orbit for each fixed parameter pair, and not a sequence of maps chosen at successive horizons. Weak convergence would imply boundedness, so it fails.

Remark. The obstruction uses nonlinearity. If T is a bounded linear contraction and $\sigma = \alpha$ with $0 < \theta < \alpha$, the same polynomial identity gives

$$x^k = p_k(T)x^0 = \frac{1}{B(\theta, \alpha-\theta)} \int_0^1 ((1-u)I + uT)^k x^0\, u^{\theta-1}(1-u)^{\alpha-\theta-1}\, \mathrm{d}u.$$

Each operator inside the integral is a contraction. Consequently $\|x^k\| \leq \|x^0\|$ for every k, so this linear subclass cannot display the amplification used above.

5.4 Finite dimensional averaged maps

The preceding counterexample has a bounded singleton fixed-point set and is firmly nonexpansive, but its space is infinite-dimensional. We now prove Theorem 6. The algorithm-specific step is a comparison estimate for the error in a Picard iteration. The subsequent vanishing-error attraction is consistent with the established perturbation theory of Reich and Zaslavski [10, 11]; we include the elementary finite-dimensional argument to make the boundedness requirement explicit.

Proposition 5.1. Let $T : H \to H$ be nonexpansive with $p \in \operatorname{Fix} T$, and suppose that $\alpha > 1$, $0 < \theta < 1$ and $\sigma > 0$. Put

$$a = \alpha - \theta > 0, \qquad c = 1 - \theta > 0, \qquad t_k = k + \sigma - 1,$$

and choose K such that $t_K > \max\{1, a, c\}$. Define z_k, w_k, r_k by (6) and set

$$J_k(p) = c\|z_k - p\|^2 + a\|w_k - p\|^2.$$

Then, for $k \geq K$,

$$J_{k+1}(p) \leq \left(1 + \frac{2c}{t_k} + \frac{(a+c)^2}{t_k^2}\right) J_k(p). \tag{36}$$

Consequently,

$$J_k(p) = O(t_k^{2c}), \qquad \|x^{k+1} - Tx^k\| = O(k^{-\theta}) \longrightarrow 0. \tag{37}$$

No boundedness assumption on the trajectory or finite-dimensional assumption is used in this proposition.

Proof. The same cancellation that gives (7) now yields

$$x^k = \left(1 - \frac{a}{t_k}\right) z_k + \frac{a}{t_k} w_k, \qquad z_{k+1} = Tx^k, \qquad w_{k+1} = \left(1 + \frac{c}{t_k}\right) w_k - \frac{c}{t_k} z_k.$$

Unlike the update in the convergence band, the last combination is not convex. Direct expansion, with $t = t_k$, gives

$$\begin{aligned} J_{k+1}(p) - J_k(p) = \frac{2ac}{t}\big(\|w_k - p\|^2 - \|z_k - p\|^2\big) + \frac{ac(a+c)}{t^2}\|r_k\|^2 \\ - c\big(\|x^k - p\|^2 - \|Tx^k - p\|^2\big). \end{aligned} \tag{38}$$

The last parenthesis is nonnegative. Also

$$ac\|r_k\|^2 \leq (a+c) J_k(p), \qquad a\|w_k - p\|^2 \leq J_k(p),$$

so (36) follows. Its product is $O(t_k^{2c})$, because the logarithm of each factor is at most $2c/t_k + (a+c)^2/t_k^2$. Finally,

$$x^{k+1} - Tx^k = -\frac{a}{t_{k+1}} r_{k+1}, \qquad \|x^{k+1} - Tx^k\|^2 \leq \frac{a(a+c)}{c\, t_{k+1}^2} J_{k+1}(p).$$

Since $c - 1 = -\theta$, this proves (37). The error controlled here is $x^{k+1} - Tx^k$, not yet the fixed-point residual $x^k - Tx^k$. ∎

Proof of Theorem 6. The case $H = \{0\}$ is immediate. Assume $\dim H \geq 1$ and write $\nu = (1 - \lambda)/\lambda > 0$. The averaged representation gives, for every $p \in S$,

$$\|Tx - p\|^2 \leq \|x - p\|^2 - \nu\|x - Tx\|^2. \tag{39}$$

We first record a radial decrease that follows from compactness and nonexpansivity. Translate one $p \in S$ to zero, and choose $R_0 > 0$ such that S lies strictly inside the ball of radius R_0. On its boundary, (39) gives

$$R_0 - \|T(R_0 u)\| > 0 \qquad (\|u\| = 1).$$

The unit sphere is compact, so the left side has a positive minimum δ. For $t \geq R_0$, nonexpansivity implies

$$\|T(tu)\| \leq \|T(R_0 u)\| + t - R_0 \leq t - \delta.$$

By Proposition 5.1, the actual trajectory satisfies $x^{k+1} = Tx^k + e_k$ with $e_k \to 0$. Eventually $\|e_k\| < \delta/2$. Its radius therefore decreases by at least $\delta/2$ whenever $\|x^k\| \geq R_0$; from inside that ball the next radius is at most $R_0 + \delta/2$. Thus the trajectory is bounded.

For completeness, attraction also follows by a compactness argument. Write $d(x) = \operatorname{dist}(x, S)$. Taking a nearest fixed point in (39) gives

$$d(Tx)^2 \leq d(x)^2 - \nu\|x - Tx\|^2.$$

Hence $d(Tx) < d(x)$ off S. On a closed ball containing the trajectory, the continuous function $d(x) - d(Tx)$ has a positive minimum η_ε on the compact set where $d(x) \geq \varepsilon/2$, if that set is nonempty. Once $\|e_k\| < \min\{\eta_\varepsilon/2, \varepsilon/2\}$, the distance decreases by at least $\eta_\varepsilon/2$ whenever it is at least $\varepsilon/2$. It must enter the smaller neighborhood in finitely many steps and remains within distance ε thereafter. Thus $d(x^k) \to 0$. Finally,

$$\|x^k - Tx^k\| \leq 2d(x^k) \longrightarrow 0.$$

If $S = \{p\}$, distance convergence is strong convergence to p. ∎

The perturbation step is a special case of established nonsummable-error results: [10, Theorems 2.1–2.2] treats sufficiently small errors under bounded-set-uniform attraction, and [11, Theorem 2.1] gives distance convergence for a bounded inexact orbit with vanishing errors. In the present finite-dimensional averaged setting, exact Picard iterates approach S uniformly on every compact initial ball. Indeed, (39) makes the exact orbit bounded with summable residual squares, so compactness and monotonicity of its distance give pointwise attraction. These distance functions are continuous and nonincreasing in the iteration index; compactness then gives uniform convergence. The radial argument

supplies the required boundedness of the inexact orbit. Neither these general robustness principles nor the compact-sphere observation is claimed as new. The comparison (37) connects them to the unchanged Fast KM recurrence.

In particular, each fixed two-dimensional block T_ε used above converges to zero under these parameters, despite the arbitrarily large transients across the block family. The single infinite direct sum in (35) has an unbounded orbit. The finite-dimensional conclusion requires bounded S and does not assert convergence to one selected point when S contains several points.

5.5 Proximal residual decay for positive parameters

We prove Theorem 7. The graph-variable formulation and mechanical-energy argument belong to the established analysis of inertial proximal algorithms with Hessian-driven damping; see [1, 2]. We use them here to identify the residual behavior of the unchanged recurrence for positive parameters, without asserting a new proximal method.

Let $f_* := \min f$, and define

$$z_k = Tx^{k-1}, \qquad g_k = \frac{x^{k-1} - z_k}{\gamma} \in \partial f(z_k) \quad (k \geq 0), \qquad v_k = z_k - z_{k-1} \quad (k \geq 1).$$

Every z_k belongs to $\operatorname{dom} f$, even if an initial point does not, and f_* is finite. Choose $K \geq 0$ with $K + \sigma > \max\{\alpha, \theta\}$. For $k \geq K$, write $n = k + \sigma$ and set

$$\mathcal{E}_k = \gamma\theta\big(f(z_{k+1}) - f_*\big) + \frac{n-\alpha}{2}\|v_{k+1}\|^2. \tag{40}$$

These quantities are finite and nonnegative. No function value at x^k is used.

Proof of Theorem 7. The original recurrence gives the exact graph relation

$$nv_{k+2} - (n-\alpha)v_{k+1} + \gamma(n-\theta)(g_{k+2} - g_{k+1}) + \gamma\theta g_{k+2} = 0. \tag{41}$$

Put

$$D_k = \langle g_{k+2}, v_{k+2}\rangle - \big(f(z_{k+2}) - f(z_{k+1})\big) \geq 0.$$

Taking the inner product of (41) with v_{k+2} and expanding the kinetic difference yields

$$\begin{aligned} \mathcal{E}_{k+1} - \mathcal{E}_k = & -\left(\alpha - \frac{1}{2}\right)\|v_{k+2}\|^2 - \frac{n-\alpha}{2}\|v_{k+2} - v_{k+1}\|^2 \\ & - \gamma(n-\theta)\langle g_{k+2} - g_{k+1}, v_{k+2}\rangle - \gamma\theta D_k. \end{aligned} \tag{42}$$

Subgradient monotonicity makes the inner product nonnegative. If $\alpha > 1/2$, telescoping gives

$$\left(\alpha - \frac{1}{2}\right)\sum_{k=K}^{\infty}\|v_{k+2}\|^2 \leq \mathcal{E}_K,$$

so $v_k \to 0$.

For $0 < \alpha \le 1/2$, use $\mathcal{E}_{k+1} \ge (n+1-\alpha)\|v_{k+2}\|^2/2$ in the same identity. This gives

$$\mathcal{E}_{k+1} \le \frac{n+1-\alpha}{n+\alpha}\mathcal{E}_k.$$

The ratio is $1 + (1-2\alpha)/(n+\alpha)$. Taking products therefore yields

$$\mathcal{E}_k = O((k+\sigma)^{1-2\alpha}), \qquad \|v_{k+1}\| = O((k+\sigma)^{-\alpha}) \longrightarrow 0. \tag{43}$$

When $\alpha < 1/2$, this argument permits growth of the energy. It gives neither a fixed objective-sublevel bound nor square summability of v_k.

To recover the actual residual, put $q_k = \gamma g_{k+1} + v_{k+1}$. Equation (41) is equivalent to

$$q_{k+1} = \left(1 - \frac{\theta}{n}\right) q_k + \frac{\theta}{n}\left(\frac{\theta-\alpha}{\theta} v_{k+1}\right).$$

On the chosen tail, the averaging weights lie in $(0,1)$ and have divergent sum, while the target vectors tend to zero. Given $\varepsilon > 0$, their norms are eventually at most ε; iteration bounds $\|q_k\|$ by ε plus a fixed initial term multiplied by a product of $1 - \theta/n$ tending to zero. Hence $q_k \to 0$, and

$$x^k - Tx^k = \gamma g_{k+1} = q_k - v_{k+1} \longrightarrow 0.$$

The original recurrence now gives

$$x^{k+1} - x^k = -\frac{\theta}{n}\gamma g_{k+1} + \left(1 - \frac{\alpha}{n}\right) v_{k+1} \longrightarrow 0.$$

No compactness or boundedness of the trajectory was used in these Hilbert-space conclusions.

For the finite-dimensional assertion, we use an elementary fact about approximate fixed points. If a nonexpansive map on $\mathbb{R}^d$ has a nonempty bounded fixed-point set, every sequence u_j with $u_j - Tu_j \to 0$ is bounded. To prove this, translate a fixed point to zero and suppose that a subsequence has $r_j = \|u_j\| \to \infty$ and $u_j/r_j \to d$, with $\|d\| = 1$. Put $e_j = u_j - Tu_j \to 0$. For every fixed y, nonexpansivity gives

$$\|Ty - u_j + e_j\|^2 \le \|y - u_j\|^2.$$

Expansion and division by $2r_j$ show that $\langle d, Ty - y\rangle \ge 0$. Taking $y = td$ gives $\langle d, T(td)\rangle \ge t$, whereas $\|T(td)\| \le t$. Thus $T(td) = td$ for every $t \ge 0$, contradicting boundedness of the fixed-point set. This proves the fact.

For $T = \operatorname{prox}_{\gamma f}$, $\operatorname{Fix} T = \operatorname{argmin} f$. When this set is $\{p\}$, residual convergence and the preceding fact give boundedness of x^k. Every cluster point is fixed by continuity of T, so finite-dimensional compactness implies $x^k \to p$. ∎

The relation to the smooth IPAHD family is concrete. On the positive shifted tail clock $q = n - \alpha$, its step-one coefficients can be chosen as

$$\beta_q = \frac{\gamma(q+\alpha-\theta)}{q}, \qquad b_q = \frac{\gamma\theta}{q}.$$

They reproduce (41), while the quantity in the hypotheses of [2, Theorem 3.1] is

$$B_q = q(b_q + \beta_q - \beta_{q+1}) - \beta_{q+1} = \gamma(\theta - 1).$$

The printed assumptions $\alpha > 1$ and $B_q > 0$ therefore exclude small α and $\theta \leq 1$. The additional ratio condition in its iterate-convergence theorem is also not met by this substitution: for $\theta > 1$, $\beta_{q+1}/B_q \to 1/(\theta - 1)$. This comparison concerns the author manuscript compiled 11 October 2021, linked in the bibliography; it does not claim that no other known argument implies the present residual result.

The conclusion for proximal maps is restricted to residual and step decay in general Hilbert space. It does not establish boundedness or weak convergence of the full trajectory there, and it does not establish convergence among multiple minimizers. In particular, no inconsistency with the infinite-dimensional unbounded-orbit result is inferred from residual decay alone.

6 Proof of Theorem 8

Proof. Write $d = \alpha - 2$ and $A = d + 1$. Define

$$w = x + t(\dot{x} + Q(x)), \qquad r = x - w. \tag{44}$$

The equation is equivalent to

$$\dot{x} = -Q(x) + \frac{w - x}{t}, \qquad \dot{w} = \frac{d}{t}(x - w), \tag{45}$$

with $w(t_0) = x_0 + t_0(v_0 + Q(x_0))$. Indeed, differentiation of (44) and substitution in (2) give $\dot{w} = -d(\dot{x} + Q(x))$; the converse follows by differentiating the first equation in (45). The vector field in (45) is globally Lipschitz in the state on each bounded time interval and has at most linear growth. The Hilbert-space Picard theorem and the usual growth estimate therefore give a unique global C^1 solution. On each compact time interval, $Q(x(\cdot))$ and $\dot{x}$ are Lipschitz. Hilbert-valued Lipschitz curves have strong derivatives almost everywhere, which gives the stated solution regularity. All energies below are locally absolutely continuous.

Fix $p \in S_Q$ and put

$$E_p(t) = d\|x(t) - p\|^2 + \|w(t) - p\|^2.$$

Direct differentiation yields

$$\dot{E}_p = -2d\langle x - p, Q(x)\rangle - \frac{2d}{t}\|r\|^2 \leq 0. \tag{46}$$

Thus x, w, r are bounded and

$$\int_{t_0}^{\infty} \frac{\|r(t)\|^2}{t}\,\mathrm{d}t \leq \frac{E_p(t_0)}{2d}. \tag{47}$$

The stronger rate follows from the same correction used for the discrete iteration. Set $c = 2A + A^2/d$ and

$$G_p(t) = t^2\|\dot{x}(t)\|^2 + cE_p(t) - A\|r(t)\|^2. \tag{48}$$

The identity

$$G_p = t^2\|\dot{x}\|^2 + 2AE_p + \frac{A}{d}\|d(x-p) + (w-p)\|^2$$

shows that $G_p \geq 0$. If $s = t\dot{x}$, equations (45) give

$$\dot{s} = -t\frac{\mathrm{d}}{\mathrm{d}t}Q(x) + \frac{A}{t}r, \qquad \dot{r} = \frac{1}{t}(s - dr).$$

Consequently, for almost every $t \geq t_0$,

$$\dot{G}_p = -2t^2\langle \dot{x}, \frac{\mathrm{d}}{\mathrm{d}t}Q(x)\rangle - 2cd\langle x - p, Q(x)\rangle - \frac{2d(c-A)}{t}\|r\|^2 \leq 0. \tag{49}$$

The first inner product is nonnegative: apply monotonicity to $x(t+h), x(t)$, divide by h^2, and pass to the time derivatives at a point where they exist. This uses no derivative of Q as an operator. The second inner product is nonnegative because $Qp = 0$.

It follows that

$$\|\dot{x}(t)\| \leq \frac{\sqrt{G_p(t_0)}}{t}, \qquad \|Q(x(t))\| \leq \frac{\sqrt{G_p(t_0)} + \sqrt{AE_p(t_0)/d}}{t}. \tag{50}$$

For the second estimate, use $Q(x) = -\dot{x} - r/t$ and $\|r\|^2 \leq AE_p/d$.

It remains to prove convergence of the trajectory. Let $q(t) = \|w(t) - p\|^2$ and $\ell_p = \lim_{t\to\infty} E_p(t)$. The exact scalar identity is

$$\dot{q} + \frac{A}{t}q = \frac{E_p(t)}{t} - \frac{d}{t}\|r(t)\|^2.$$

Multiplication by t^A and integration give

$$q(t) = \left(\frac{t_0}{t}\right)^A q(t_0) + \frac{1}{t^A}\int_{t_0}^{t} u^{A-1}E_p(u)\,\mathrm{d}u - \int_{t_0}^{t}\left(\frac{u}{t}\right)^A \frac{d\|r(u)\|^2}{u}\,\mathrm{d}u.$$

The middle term tends to ℓ_p/A: replace $E_p(u)$ by ℓ_p and bound the error after a fixed large time by its tail supremum. The last term tends to zero by (47). Indeed, its integral over a fixed finite interval tends to zero, while the remaining integral is bounded by an arbitrarily small L^1 tail. Therefore $q(t) \to \ell_p/A$. Hence

$$\|x(t) - p\|^2 = \frac{E_p(t) - q(t)}{d} \longrightarrow \frac{\ell_p}{A}.$$

Every weak sequential cluster point $\bar{x}$ is a zero of Q. To verify this directly, combine $Q(x(t)) \to 0$ with monotonicity to obtain $\langle Qy, y - \bar{x}\rangle \geq 0$ for every $y \in H$. Set $y = \bar{x} - \varepsilon Q\bar{x}$, divide by $\varepsilon > 0$ and let $\varepsilon \downarrow 0$. Continuity gives $\|Q\bar{x}\|^2 \leq 0$. The distance limits and the Opial argument used in the discrete proof show that there is only one weak

cluster point. To pass from sequences to the full trajectory, fix $h \in H$. If $\langle x(t) - \bar{x}, h\rangle$ did not tend to zero, a sequence of times tending to infinity would stay separated from zero. Boundedness supplies a weakly convergent subsequence, whose limit must be $\bar{x}$, a contradiction. This proves weak convergence of the trajectory; finite-dimensional strong convergence follows. ∎

The orders in Theorem 8 cannot be replaced by universal little-$o(1/t)$ estimates. The example in [5, Eq. (2.18)], with $H = \mathbb{R}^2 \cong \mathbb{C}$, $Qx = ix$, $\alpha = 3$ and

$$x(t) = \frac{1 + it}{t^2} e^{-it},$$

satisfies

$$t^2 \|Q(x(t))\|^2 = 1 + t^{-2}, \qquad t^2 \|\dot{x}(t)\|^2 = 1 + 4t^{-4}.$$

Thus both scaled norms tend to 1. This continuous obstruction is already present in the cited work; Theorem 8 supplies the matching upper order and weak convergence for general monotone Lipschitz mappings.

7 Scaled limits and weighted summability

The preceding energies give more than bounded scaled norms. They also show that the residual and velocity have the same asymptotic magnitude at the endpoint.

Proposition 7.1. Under the assumptions of Theorem 1, with $\alpha > 2$ and $\theta = \alpha - 1$, the discrepancy r_k in (6) converges strongly to zero, and there is a finite number $L \geq 0$ such that

$$k\|x^k - Tx^k\| \longrightarrow L, \qquad k\|x^{k+1} - x^k\| \longrightarrow L.$$

For every $p \in \operatorname{Fix} T$,

$$L^2 = \lim_{k\to\infty} G_k(p) - c \lim_{k\to\infty} E_k(p), \qquad c = 2A + A^2/d.$$

Under the assumptions of Theorem 8, the continuous discrepancy $r(t)$ in (44) also converges strongly to zero. There is a finite number $L \geq 0$, depending on that trajectory, such that

$$t\|Q(x(t))\| \longrightarrow L, \qquad t\|\dot{x}(t)\| \longrightarrow L,$$

and $L^2 = \lim_{t\to\infty} G_p(t) - c \lim_{t\to\infty} E_p(t)$ for every $p \in S_Q$.

Proof. Here $a = 1$, $b = d = \alpha - 2$ and $A = d + 1$, as in Section 3.1. We first record an elementary decay observation. If a nonnegative sequence h_k satisfies

$$|h_{k+1} - h_k| \leq C/t_k, \qquad \sum_{k\geq K} h_k^2/t_k < \infty,$$

then $h_k \to 0$. For $C = 0$ this is immediate. Otherwise, suppose that $h_n \geq \varepsilon$ for infinitely many indices. Set $\delta = \varepsilon/(4C)$ and choose each such n sufficiently large that $1/t_n \leq \delta$. Let $m > n$ be the first index with $\sum_{j=n}^{m-1} 1/t_j \geq \delta$. For $n \leq k < m$, the preceding partial sum is less than δ, so $h_k \geq 3\varepsilon/4$. Thus the weighted square sum on this block is at least $9\varepsilon^2\delta/16$. Infinitely many disjoint such blocks contradict the assumed summability.

The update (22) gives

$$\|r_{k+1} - r_k\| \leq \|x^k - x^{k-1}\| + \frac{d}{t_k}\|r_k\| \leq \frac{B + dM}{t_k},$$

where $B = \sup\|s_k\|$ and $M = \sup\|r_k\|$ are finite. Apply the observation to $h_k = \|r_k\|$, using (10), to obtain $r_k \to 0$. Since $E_k(p)$ and $G_k(p)$ have limits,

$$\|s_k\|^2 = G_k(p) - cE_k(p) + A\|r_k\|^2$$

also has a finite limit $L^2 \geq 0$. The exact residual identity

$$t_k(x^k - Tx^k) = -s_{k+1} + \left(\frac{A}{t_k} - 1\right) r_k$$

shows that its norm has the same limit. Finally $t_k/k \to 1$ and $t_{k+1}/k \to 1$ give the stated indexing.

For the continuous problem, boundedness of $t\dot{x}$ and r gives $\|\dot{r}(t)\| \leq C/t$. In logarithmic time $\tau = \log(t/t_0)$, the curve $R(\tau) = r(t_0 e^\tau)$ is Lipschitz and satisfies

$$\int_0^\infty \|R(\tau)\|^2 \, \mathrm{d}\tau < \infty.$$

A Lipschitz square-integrable curve converges to zero: otherwise disjoint intervals of one fixed positive length would each contribute a fixed positive amount to this integral. Hence $r(t) \to 0$. The identity $t^2\|\dot{x}\|^2 = G_p - cE_p + A\|r\|^2$ and monotonicity of the two energies prove convergence of the scaled velocity norm. The identity $tQ(x) = -t\dot{x} - r$ gives the same limit for the residual norm. ∎

7.1 A fixed firmly nonexpansive example

The existence of a vanishing scaled residual does not imply the weighted square summability that is often used to prove little-o rates. This distinction remains relevant even for firmly nonexpansive maps. The following example uses the standard direct sum of averaged rotations appearing in Davis and Yin [7]; the calculation below concerns the endpoint polynomial generated by (1), rather than ordinary powers of that operator.

Recall that T is firmly nonexpansive when

$$\|Tx - Ty\|^2 \leq \langle Tx - Ty, x - y\rangle \qquad (x, y \in H).$$

Every average $(I + R)/2$ with R nonexpansive has this property.

Proposition 7.2. There exist a real Hilbert space H, a firmly nonexpansive linear map $T : H \to H$ with $\operatorname{Fix} T = \{0\}$, and an initial point $x^0 \in H$ such that the iteration (1), with $\alpha = \sigma = 3$ and $\theta = 2$, satisfies

$$x^k \longrightarrow 0, \qquad k\|x^k - Tx^k\| \longrightarrow 0, \qquad \sum_{k=1}^{\infty} k\|x^k - Tx^k\|^2 = \infty.$$

The map and the initial point are fixed throughout the iteration. The choice of x^{-1} is arbitrary.

Proof. We first calculate a scalar orbit. For $k \geq 0$, put

$$p_k(z) = \frac{2}{(k+1)(k+2)} \sum_{l=0}^{k} (l+1) z^l.$$

These polynomials satisfy $p_0 = 1$, $p_1 = (1+2z)/3$ and

$$(k+3)p_{k+1}(z) = \big(k+1+(k+2)z\big)p_k(z) - kzp_{k-1}(z).$$

This is precisely (1) for the scalar linear map $Tz = \lambda z$. Summing the geometric series and its derivative gives

$$(1-z)p_k(z) = -\frac{2z^{k+1}}{k+2} + \frac{2(1-z^{k+1})}{(k+1)(k+2)(1-z)}. \tag{51}$$

Let $0 < \delta \leq 1/128$ and $z = (1 + e^{2i \arcsin \delta})/2$. Then $|1-z| = \delta$ and $|z|^2 = 1 - \delta^2$. Define

$$m = \left\lceil \frac{16}{\delta} \right\rceil, \qquad M = \left\lfloor \frac{1}{4\delta^2} \right\rfloor.$$

For $m \leq k \leq M$, Bernoulli's inequality gives

$$|z|^{2(k+1)} \geq 1 - (k+1)\delta^2 \geq \frac{1}{2}.$$

The second term of (51) has absolute value at most $1/[4(k+2)]$, since $(k+1)\delta \geq 16$. Therefore

$$|(1-z)p_k(z)| \geq \frac{\sqrt{2} - 1/4}{k+2} \geq \frac{1}{k+2}.$$

It follows that

$$\begin{aligned} \sum_{k=m}^{M} k|(1-z)p_k(z)|^2 &\geq \frac{1}{4} \sum_{k=m}^{M} \frac{1}{k} \geq \frac{1}{4} \log \frac{M+1}{m} \\ &\geq \frac{1}{4} \log \frac{1}{68\delta}. \end{aligned} \tag{52}$$

Here $m \leq 17/\delta$ and $M + 1 > 1/(4\delta^2)$; the chosen range of δ ensures $2 \leq m \leq M$.

Now take $H = \ell^2(\mathbb{N};\mathbb{C})$, regarded as a real Hilbert space, and for $j \geq 1$ set

$$\delta_j = e^{-(j+8)^2}, \qquad \lambda_j = \frac{1 + e^{2i \arcsin \delta_j}}{2}, \qquad (Tx)_j = \lambda_j x_j, \qquad x_j^0 = \frac{1}{j}.$$

The operator T is the average of the identity and a diagonal unitary map, so it is firmly nonexpansive. Each $\lambda_j \neq 1$, hence $\operatorname{Fix} T = \{0\}$. Moreover, $\|x^0\|^2 = \sum_{j\geq 1} j^{-2} < \infty$. The first momentum coefficient vanishes, and the polynomial recurrence proves

$$x_j^k = \frac{p_k(\lambda_j)}{j}.$$

All coefficients of p_k are nonnegative and sum to one. Thus $|p_k(\lambda_j)| \leq 1$, while $p_k(\lambda_j) \to 0$ for every fixed j, since $|\lambda_j| < 1$. Dominated convergence gives $x^k \to 0$ in norm.

Abel summation also gives $k|(1-z)p_k(z)| \leq 4$ for $|z| \leq 1$. Indeed, the coefficients of p_k increase and their largest value is $2/(k+2)$, so the sum of the absolute coefficients of $(1-z)p_k(z)$ is $4/(k+2)$. For each fixed $|z| < 1$, (51) gives $k|(1-z)p_k(z)| \to 0$. Dominated convergence, with summable majorant $16/j^2$, therefore proves $k\|x^k - Tx^k\| \to 0$.

Finally Tonelli's theorem and (52) yield

$$\begin{aligned}\sum_{k=1}^{\infty} k\|x^k - Tx^k\|^2 &= \sum_{j=1}^{\infty} \frac{1}{j^2} \sum_{k=1}^{\infty} k|(1-\lambda_j)p_k(\lambda_j)|^2 \\ &\geq \frac{1}{4} \sum_{j=1}^{\infty} \frac{(j+8)^2 - \log 68}{j^2} = \infty.\end{aligned}$$

This proves all the assertions for one fixed map and one fixed initial vector. ∎

The same map can be realized as a Douglas–Rachford map for two closed subspaces: in coordinate j, take the real axis and the line making angle $\arcsin \delta_j$ with it. Their reflected composition is multiplication by $e^{2i \arcsin \delta_j}$. This is the subspace construction in [7]; only its endpoint-iteration behavior is used here.

For completeness, this same fixed example has no power improvement over $1/k$. At $m_j = \lceil 16/\delta_j \rceil$, the scalar bound gives

$$m_j \|x^{m_j} - Tx^{m_j}\| \geq \frac{m_j}{j(m_j+2)} \geq \frac{1}{2j}.$$

Since $m_j \geq 16e^{(j+8)^2}$, for every $\varepsilon > 0$ we have

$$\limsup_{k\to\infty} k^{1+\varepsilon} \|x^k - Tx^k\| = \infty.$$

This is consistent with the little-$o(1/k)$ limit already proved for the example.

At the acceleration boundary, the results above do not settle whether the scaled limit in Proposition 7.1 is zero for every strictly averaged nonlinear map $T = (1-\lambda)I + \lambda R$, where $0 < \lambda < 1$ and R is nonexpansive. Proposition 7.2 rules out weighted residual-square summability as a general route to that conclusion; it does not provide a counterexample to the little-$o(1/k)$ conclusion itself.

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